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Progress report #1

Due date: 02/18/26

Company A

1. LAN Requirements

Company A has 230 employees divided into four departments:

- Engineering – 100 employees
- Manufacturing – 80 employees
- Sales/Marketing – 30 employees
- Administration – 20 employees

The LAN must include the following servers:

- DHCP server (for automatic IP assignment)
- DNS server (for name resolution)
- FTP Server 1 (for Engineering department only)
- FTP Server 2 (for other departments)
- Web server
- Email server

Company A is assigned the IP address range:

216.244.168.0 – 216.244.175.255

The IP address 216.244.175.254 is reserved for the WAN endpoint.

VLSM is used to efficiently divide the IP address space based on department size.

The LAN must meet the following requirements:

- The network should be connected to the internet through a firewall.
- The web server and the email server should be deployed in DMZ.

- A VPN should be configured to allow remote users to securely connect to the LAN using IPsec or SSL.
- A guest WLAN should be configured so visitors can also access only web servers.

2. IP Address Allocation Using VLSM

Engineering Department (100 Employees)

- Network ID: 216.244.168.0
- Subnet Mask: 255.255.255.128
- CIDR: /25
- Broadcast Address: 216.244.168.127
- First Host Address: 216.244.168.1
- Last Host Address: 216.244.168.126
- Maximum Hosts: 126

Manufacturing Department (80 Employees)

- Network ID: 216.244.168.128
- Subnet Mask: 255.255.255.128
- CIDR: /25
- Broadcast Address: 216.244.168.255
- First Host Address: 216.244.168.129
- Last Host Address: 216.244.168.254
- Maximum Hosts: 126

Sales/Marketing Department (30 Employees)

- Network ID: 216.244.169.0
- Subnet Mask: 255.255.255.224
- CIDR: /27
- Broadcast Address: 216.244.169.31
- First Host Address: 216.244.169.1
- Last Host Address: 216.244.169.30
- Maximum Hosts: 30

Administration Department (20 Employees)

- Network ID: 216.244.169.32
- Subnet Mask: 255.255.255.224
- CIDR: /27
- Broadcast Address: 216.244.169.63
- First Host Address: 216.244.169.33
- Last Host Address: 216.244.169.62

- Maximum Hosts: 30

Internal Server Subnet

(DHCP, DNS, FTP1, FTP2)

- Network ID: 216.244.169.64
- Subnet Mask: 255.255.255.240
- CIDR: /28
- Broadcast Address: 216.244.169.79
- First Host Address: 216.244.169.65
- Last Host Address: 216.244.169.78
- Maximum Hosts: 14

DMZ Subnet

(Web and Email Servers)

- Network ID: 216.244.169.80
- Subnet Mask: 255.255.255.248
- CIDR: /29
- Broadcast Address: 216.244.169.87
- First Host Address: 216.244.169.81
- Last Host Address: 216.244.169.86
- Maximum Hosts: 6

Guest WLAN Subnet

- Network ID: 216.244.169.128
- Subnet Mask: 255.255.255.192
- CIDR: /26
- Broadcast Address: 216.244.169.191
- First Host Address: 216.244.169.129
- Last Host Address: 216.244.169.190
- Maximum Hosts: 62

WAN Subnet

- Network ID: 216.244.175.252
- Subnet Mask: 255.255.255.252
- CIDR: /30
- Broadcast Address: 216.244.175.255
- First Host Address: 216.244.175.253
- Last Host Address: 216.244.175.254
- Maximum Hosts: 2

